

# Veronica Boyce

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## EDUCATION

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**Stanford University**

**Stanford, CA**

Doctor of Philosophy in Psychology

September 2025

Dissertation: *Constraints on the emergence and interpretability of linguistic conventions*

**Massachusetts Institute of Technology**

**Cambridge, MA**

Bachelor of Science in Brain & Cognitive Science

June 2019

## EMPLOYMENT

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**MIT, Computational Psycholinguistics Lab**

**Cambridge, MA**

NSF SBE Postdoctoral Research Fellow

Jan 2026 – present

**Stanford University, Language and Cognition Lab**

**Stanford, CA**

Postdoctoral Researcher

Sep 2025 – Jan 2026

**MIT, Computational Psycholinguistics Lab**

**Cambridge, MA**

Research Support Associate

June 2019 – July 2020

## AWARDS, FELLOWSHIPS, & GRANTS

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NSF SBE Postdoctoral Research Fellowship

Jan 2026 – Jan 2028

Open Science Innovator Prize, *Stanford Center for Open and Reproducible Science*

2025

The Albert H. and Barbara R. Hastorf Prize for Teaching, *Stanford Department of Psychology*

2025

## PUBLICATIONS

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M. C. Frank, V. A. Marchman, C. A. Bergey, **V. Boyce**, M. Braginsky, G. Kachergis, J. Manke-witz, S. C. Meylan, B. Prystawski, N. Ram, R. Z. Sparks, A. Steffan, A. W. M. Tan, and M. Zettersten (2026). “Continuous developmental changes in word recognition support language learning across early childhood”. *eLife*. [elifesciences.org/articles/109636](https://elifesciences.org/articles/109636).

**V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2024). “Interaction Structure Constrains the Emergence of Conventions in Group Communication”. *PNAS*. [doi.org/10.1073/pnas.2403888121](https://doi.org/10.1073/pnas.2403888121).

**V. Boyce**, B. Prystawski, A. B. Abutto, E. M. Chen, Z. Chen, H. Chiu, I. Ergin, A. Gupta, C. Hu, B. Kemmann, N. Klevak, V. Y. Q. Lua, M. Mazzaferro, K. Mon, D. Ogunbamowo, A. Pereira, J. Troutman, S. Tung, R. Uricher, and M. C. Frank (2024). “Estimating the replicability of psychology experiments after an initial failure to replicate”. *Collabra: Psychology*. [doi.org/10.1525/collabra.125685](https://doi.org/10.1525/collabra.125685).

A. Chuey, **V. Boyce**, C. Anjie, and M. C. Frank (2024). “Conducting Developmental Research Online vs. in-Person: A Meta-Analysis”. *Open Mind*. [doi.org/10.1162/opmi\\_a\\_00147](https://doi.org/10.1162/opmi_a_00147).

- M. Soderstrom, J. Rocha-Hidalgo, L. E. Muñoz, A. Bochynska, J. F. Werker, B. Skarabela, A. Seidl, Y. Ryjova, J. L. Rennels, C. E. Potter, M. Paulus, M. Ota, N. M. Olesen, K. M. Nave, J. Mayor, A. Martin, L. C. Machon, C. Lew-Williams, E.-S. Ko, H. Kim, N. Kartushina, M. Kammermeier, A. Jessop, J. F. Hay, N. Havron, E. E. Hannon, J. K. Hamlin, N. Gonzalez-Gomez, A. Gampe, T. Fritzsche, M. C. Frank, S. Durrant, C. Davies, C. Cashon, K. Byers-Heinlein, **V. Boyce**, A. K. Black, C. Bergmann, L. Anderson, M. K. Alshakhori, A. H. Al-Hoorie, and A. S. M. Tsui (2024). “Testing the relationship between preferences for infant-directed speech and vocabulary development: A multi-lab study”. *Journal of Child Language*. [doi.org/10.1017/S0305000924000254](https://doi.org/10.1017/S0305000924000254).
- V. Boyce** and R. P. Levy (2023). “A-Maze of Natural Stories: Comprehension and Surprisal in the Maze Task”. *Glossa Psycholinguistics*. [escholarship.org/uc/item/6vh9d8zm](https://escholarship.org/uc/item/6vh9d8zm).
- V. Boyce**, M. Mathur, and M. C. Frank (2023). “Eleven Years of Student Replication Projects Provide Evidence on the Correlates of Replicability in Psychology”. *Royal Society Open Science*. [doi.org/10.1098/rsos.231240](https://doi.org/10.1098/rsos.231240).
- M. Zettersten, D. Yurovsky, T. L. Xu, S. Uner, A. S. M. Tsui, R. M. Schneider, A. N. Saleh, S. C. Meylan, V. A. Marchman, J. Mankewitz, K. MacDonald, B. Long, M. Lewis, G. Kachergis, K. Handa, B. deMayo, A. Carstensen, M. Braginsky, **V. Boyce**, N. S. Bhatt, C. A. Bergey, and M. C. Frank (2023). “Peekbank: An Open, Large-Scale Repository for Developmental Eye-Tracking Data of Children’s Word Recognition”. *Behavior Research Methods*. [doi.org/10.3758/s13428-022-01906-4](https://doi.org/10.3758/s13428-022-01906-4).
- V. Boyce**, R. Futrell, and R. P. Levy (2020). “Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty”. *Journal of Memory and Language*. [doi.org/10.1016/j.jml.2019.104082](https://doi.org/10.1016/j.jml.2019.104082).

## CONFERENCE PROCEEDINGS

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- A. W. M. Tan, B. Prystawski, and **V. Boyce** (2026). “Which one is banana man? Evaluating vision-language models in multi-turn pragmatic interpretation”. *To appear in Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. Preprint at [arxiv.org/abs/2608.29571](https://arxiv.org/abs/2608.29571).
- V. Boyce**, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). “Idiosyncratic but not opaque: Linguistic conventions formed in reference games are interpretable by naïve humans and vision-language models”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, [escholarship.org/uc/item/16c4n85d](https://escholarship.org/uc/item/16c4n85d).
- V. Boyce**, R. Z. Sparks, Y. Mofor, and M. C. Frank (2025). “Preschoolers can form conventional pacts with each other to communicate about novel referents”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, [osf.io/dpgfw\\_v1](https://osf.io/dpgfw_v1).
- A. W. M. Tan, B. Prystawski, **V. Boyce**, and M. C. Frank (2025). “Context informs pragmatic interpretation in vision-language models”. *CogInterp workshop NeurIPS 2025*. Poster, preprint at [arxiv.org/abs/2511.03908](https://arxiv.org/abs/2511.03908).
- V. Boyce** and M. C. Frank (2023). “Communicative Reduction in Referring Expressions within a Multi-Player Negotiation Game”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, [escholarship.org/uc/item/8dq8c2s6](https://escholarship.org/uc/item/8dq8c2s6).
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2022). “Two’s Company but Six Is a Crowd: Emergence of Conventions in Multiparty Communication Games”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Talk, [osf.io/preprints/psyarxiv/xd7ep](https://osf.io/preprints/psyarxiv/xd7ep).

- J. Mankewitz, **V. Boyce**, B. Waldon, G. Loukatou, D. Yu, J. Mu, N. D. Goodman, and M. C. Frank (2021). “Multi-Party Referential Communication in Complex Strategic Games”. *Meaning in Context: Pragmatic Communication in Humans and Machines workshop. NeurIPS*. PsyArXiv. Talk, [osf.io/preprints/psyarxiv/tfb3d](https://osf.io/preprints/psyarxiv/tfb3d).
- M. Zettersten, C. A. Bergey, N. S. Bhatt, **V. Boyce**, M. Braginsky, A. Carstensen, B. deMayo, G. Kachergis, M. Lewis, B. Long, K. MacDonald, J. Mankewitz, S. C. Meylan, A. N. Saleh, R. M. Schneider, A. S. M. Tsui, S. Uner, T. L. Xu, D. Yurovsky, and M. C. Frank (2021). “Peekbank: Exploring Children’s Word Recognition through an Open, Large-Scale Repository for Developmental Eye-Tracking Data”. *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, [escholarship.org/uc/item/9j05h4n1](https://escholarship.org/uc/item/9j05h4n1).
- V. Boyce**, A. Aravind, and M. Hackl (2017). “Lexical and Syntactic Effects on Auxiliary Selection: Evidence from Child French”. *Boston University Conference on Language Development*. Talk, [lingref.com/buclid/41/BUCLD41-08.pdf](https://lingref.com/buclid/41/BUCLD41-08.pdf).

## PREPRINTS (UNDER REVIEW)

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- V. Boyce**, V. A. Marchman, H. A. Baumgartner, C. A. Bergey, M. Braginsky, F. Giovannetti, G. Kachergis, J. Mankewitz, S. C. Meylan, B. Prystawski, R. Z. Sparks, A. Steffan, A. W. M. Tan, M. C. Frank, and M. Zettersten (2026). “Data-driven recommendations for increasing reliability and validity in measures of infant word recognition”. Preprint, under review, [osf.io/preprints/psyarxiv/28gq4\\_v2](https://osf.io/preprints/psyarxiv/28gq4_v2).
- V. Boyce**, R. Z. Sparks, and M. C. Frank (2026). “Preschoolers can coordinate with each other to communicate about novel referents”. Preprint, under review, [osf.io/preprints/psyarxiv/t76a5\\_v2](https://osf.io/preprints/psyarxiv/t76a5_v2).
- V. Boyce**, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, M. C. Frank, and R. D. Hawkins (2026). “Refbank: an open repository of iterated reference game data”. Preprint, under review, [osf.io/preprints/psyarxiv/d93gy\\_v1](https://osf.io/preprints/psyarxiv/d93gy_v1).
- A. N. Meltzoff, S. Farwaha, V. Slaughter, E. A. Simpson, R. Brooks, M. Nielsen, H. A. Baumgartner, L. Bayet, S. Benavides-Varela, **V. Boyce**, K. Byers-Heinlein, J. Döncke, B. P. Flaherty, S. H. Forbes, M. C. Frank, A. Geraci, K. Hamlin, M. Heimann, N. Keven, J. Lany, J. R. Lelonkiewicz, T. S. Leung, S. Maylott, D. Moreau, E. Quadrelli, P. J. Reschke, C. Turati, S. Vincini, S. M. Weiss, T. S. Zamuner, and G. Zeng (2026). “ManyBabies 6: A Multi-Lab Investigation of Neonatal and Early Imitation”. Stage 1 Registered Report, under review, pre-print, [osf.io/preprints/psyarxiv/fe7pj\\_v1](https://osf.io/preprints/psyarxiv/fe7pj_v1).
- V. Boyce**, A. Acharya, and M. C. Frank (2025). “Use of communication channels in 2x2 strategic games”. White paper, [osf.io/preprints/psyarxiv/y6ahw\\_v1](https://osf.io/preprints/psyarxiv/y6ahw_v1).
- L. Röseler, L. Wallrich, H. Hartmann, L. Altegoer, **V. Boyce**, S. M. Field, J. Goltermann, J. Hüffmeier, C. R. Pennington, M.-M. Pittelkow, P. Silverstein, D. van Ravenzwaaij, and F. Azevedo (2025). “Handbook on Conducting Reproduction and Replication Studies”. Preprint, under revision following MetaROR peer review, [doi.org/10.5281/zenodo.16990114](https://doi.org/10.5281/zenodo.16990114).

## CONFERENCE PRESENTATIONS

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- V. Boyce**, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, B. Prystawski, M. C. Frank, and R. D. Hawkins (2026). “Refbank: an open repository of iterated reference games”. The 39th Annual Conference on Human Sentence Processing. Poster.
- V. Boyce** and R. P. Levy (2025). “Linguistic, but not visual, discourse context modulates syntactic parsing in the Maze”. The 38th Annual Conference on Human Sentence Processing. Poster.
- V. Boyce**, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). “Context matters: History and informativity affect understanding of opaque references”. The 38th Annual Conference on Human Sentence Processing. Poster.
- J. Vigly, **V. Boyce**, M. Socolof, J. Michaelov, T. von der Malsburg, and R. P. Levy (2025). “Implicit gender bias in linguistic descriptions for expected events: The case of the 2024 United States presidential election”. The 38th Annual Conference on Human Sentence Processing. Poster.
- V. Boyce**, I. Chen, R. Z. Sparks, M. Perez, and M. C. Frank (2024). “4-5 year old children can successfully communicate using ad-hoc referential expressions”. CAMP 6. Talk.
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). “Thick Feedback Facilitates Referential Coordination in Larger Groups”. CAMP 5. Talk.
- V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). “Thick Feedback Facilitates Referential Coordination in Larger Groups”. HSP. Poster.
- T. Poppels, **V. Boyce**, C. Ajunwa, T. von der Malsburg, and R. P. Levy (2021). “Bias against ‘she’ Pronouns Can Be Rapidly Overcome by Changing Event Expectations”. CUNY Human Sentence Processing Conference. Short talk.
- V. Boyce** and R. P. Levy (2020). “A-Maze of Natural Stories: Texts Are Comprehensible Using the Maze Task”. Architectures and Mechanisms for Language Processing. Talk.
- V. Boyce**, R. Futrell, and R. P. Levy (2019). “Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty”. CUNY Human Sentence Processing Conference. Poster.
- V. Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). “Female Gender Is Consistently Under-Expressed in Pronoun Production and under-Inferred in Comprehension”. Linguistic Society of American Annual Meeting. Poster.
- V. Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). “Remember ‘Him’, Forget ‘Her’: Gender Bias in the Comprehension of Pronominal Referents.” CUNY Human Sentence Processing Conference. Poster.
- V. Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2018). “Implicit Gender Biases in the Production and Comprehension of Pronominal References”. CUNY Human Sentence Processing Conference. Poster.

## INVITED TALKS

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|                                                     |            |
|-----------------------------------------------------|------------|
| Harvard LangCog Talk Series, Harvard University     | Sep 2026   |
| Goldenberg Lab, Harvard Business School             | Nov 2025   |
| Language and Cognitive Development Lab, UC Berkeley | Oct 2025   |
| Saxelab, MIT                                        | Feb 2023   |
| QuantLang Lab, UC Irvine                            | April 2021 |

## TEACHING

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|---------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Stanford, Department of Psychology</b>                                                               | <b>Stanford, CA</b>      |
| TA for PSYCH 125: Research Methods in Psychology                                                        | Jan 2024 – March 2024    |
| <i>First offering of course; designed and led course replication projects with bi-weekly milestones</i> |                          |
| TA for PSYCH 251: Experimental Methods                                                                  | Sep 2023 – Dec 2023      |
| <i>Designed and supervised re-replication option for course project</i>                                 |                          |
| <i>Led collective writing of joint paper reporting findings across projects (see Publications)</i>      |                          |
| TA for PSYCH 251: Experimental Methods                                                                  | Sep 2022 – Dec 2022      |
| Guest Lecturer, PSYCH 251: Experimental Methods                                                         | 2023, 2025               |
| <i>Topic: meta-analysis and meta-science</i>                                                            |                          |
| TA for PSYCH 45: Introduction to Learning and Memory                                                    | April 2023 – June 2023   |
| TA for PSYCH 220B: Probabilistic Models of Cognition: Language                                          | Sep 2021 – Dec 2021      |
| <b>MIT, Department of Brain &amp; Cognitive Science</b>                                                 | <b>Cambridge, MA</b>     |
| TA for Computational Psycholinguistics                                                                  | February 2019 – May 2019 |

## MENTORING

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|----------------------------------------------------------------------------------------------|-------------|
| Benji Fernandez (Stanford, undergraduate honors thesis in progress, co-advisor)              | 2025 – 2026 |
| <i>Co-author, poster at HSP and manuscript, see Conference Presentations &amp; Preprints</i> |             |
| Fiona Feng (Queens College undergraduate research assistant)                                 | 2025 – 2026 |
| <i>Co-author, poster at HSP and manuscript, see Conference Presentations &amp; Preprints</i> |             |
| Ryder Fried (Stanford undergraduate research assistant)                                      | 2025        |
| <i>Co-author, poster at HSP and manuscript, see Conference Presentations &amp; Preprints</i> |             |
| Spurti Nimbali (Stanford undergraduate research assistant)                                   | 2025        |
| Zihan Wang (Stanford undergraduate research assistant)                                       | 2025        |
| Yannick Mofor (Stanford undergraduate research assistant)                                    | 2024        |
| <i>Co-author, CogSci proceedings, see Conference Proceedings</i>                             |             |
| Rebecca Pizzitola (Stanford undergrad)                                                       | 2024        |
| Ilaria Chen (Stanford undergraduate research assistant)                                      | 2023        |
| <i>Co-author, talk at CAMP, see Conference Presentations</i>                                 |             |
| Malia Perez (Stanford undergraduate research assistant)                                      | 2023        |
| <i>Co-author, talk at CAMP, see Conference Presentations</i>                                 |             |
| <i>Now PhD student at UNC Chapel Hill</i>                                                    |             |
| Brittany Linus (Stanford undergraduate research assistant)                                   | 2022        |
| Chelsea Ajunwa (MIT undergraduate research assistant)                                        | 2020        |
| <i>Co-author, short talk at HSP, see Conference Presentations</i>                            |             |
| <i>Now PhD student at Northeastern University</i>                                            |             |
| Arun Wongprommoon (MIT undergraduate research assistant)                                     | 2019        |

## AD HOC REVIEWING

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*Journals:* Replication Research (2026); Journal of Personality and Social Psychology: Attitudes and Social Cognition (2026); Infant Behavior and Development; Open Mind (2025); Language, Cognition, and Neuroscience (2024); Royal Society Open Science (2024); Journal of Experimental Psychology: Learning, Memory, and Cognition (2023); Journal of Memory and Language (2023);

Glossa Psycholinguistics (2021)

*Co-reviewed (with Michael C. Frank):* Psychological Science (2025), Perspectives on Psychological Science (2025), Nature Human Behavior (2024), Glossa Psycholinguistics (2023)

*Conferences:* CogSci (2021–2026), HSP (2026), California Meeting of Psycholinguists (2023), Society for Computation in Linguistics (2020–2026), Association for Computational Linguistics (2019)

## REFERENCES

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Dr. Roger Levy, Department of Brain and Cognitive Sciences, MIT, rplevy@mit.edu

Dr. Michael C. Frank, Department of Psychology, Stanford University, mcfrank@stanford.edu

Dr. Robert Hawkins, Department of Linguistics, Stanford University, rdhawkins@stanford.edu